MULTI-ASSET RESEARCH

COMMODITIES & INFLATION PROTECTION

Asset Class Deep Dive & Multi-Asset Strategy Framework

Updated Through YE 2025

EXECUTIVE SUMMARY

Commodities & Inflation Protection — Investment Thesis

Diversification Imperative

Stock-bond correlation changes have hurt the traditional diversification model. Commodities (-0.27 bond correlation) and gold (conditional negative equity correlation during stress) provide critical portfolio diversification.

Inflation Hedge Validated

S&P GSCI returned +586% during 1970s stagflation. Commodities rise 7-9% per 1% of unexpected inflation — the highest inflation beta of any major asset class.

Structural Tailwinds

Copper structural deficit (80-100 kg per EV), gold central bank buying (755-800 tonnes/yr), silver industrial demand from solar create multi-year commodity demand tailwinds.

Multi-Asset Inflation Shield

Combining broad commodities, gold, TIPS, REITs, and commodity equities creates an all-weather inflation protection sleeve adapting to rising inflation, stagflation, and deflation.

AGENDA

01	What Are Commodities? — Definition, Classification & Market Structure
02	Evolution of the Asset Class — History, Growth & Index Construction
03	Historical Performance — Returns by Decade & Market Regime
04	Correlation & Diversification — Quantitative Analysis
05	Individual Commodity Profiles — Deep Dives on Major Commodities
06	Fundamental Outlook — Supply-Demand & Structural Tailwinds
07	TIPS Primer — Mechanics, Yields & Inflation Protection
08	REITs Primer — Real Estate as an Inflation Hedge
09	Multi-Asset Inflation Strategy — Four Regime-Based Portfolios
10	Scenario Analysis, Implementation & Recommendation

01

What Are Commodities?

Definition, Classification & Market Structure

WHAT ARE COMMODITIES?

Standardized raw materials traded on global exchanges — the building blocks of the economy

- Commodities are physical raw materials that serve as inputs to industrial production, energy generation, and food supply.
- They are standardized and fungible — one unit is interchangeable with another of the same grade.
- Investors access them through futures, ETFs, or equities of commodity producers.

ENERGY 57% GSCI

Crude Oil, Natural Gas, Gasoline, Heating Oil

PRECIOUS METALS 19% GSCI

Gold, Silver, Platinum, Palladium

INDUSTRIAL METALS 16% GSCI

Copper, Aluminum, Zinc, Nickel, Lead

AGRICULTURE 21% GSCI

Corn, Wheat, Soybeans, Cotton, Coffee

LIVESTOCK 9% GSCI

Live Cattle, Lean Hogs, Feeder Cattle

Source: S&P GSCI Index Methodology, 2025. GSCI weights shown.

COMMODITY MARKET SIZE & GROWTH

A $2.4 trillion derivatives market with rapidly growing investor access

$2.4T	137B	$137B+	25
OTC Derivatives Notional (2024)	Futures Contracts Traded (2023)	Commodity ETF/ETP Assets (2022)	Contracts in BCOM Index

Institutional Adoption Accelerating

Commodity futures trading volume reached a record 137.3 billion contracts in 2023 (+64% YoY).
ICE alone traded 2 billion contracts in 2024-25, with oil futures reaching 655 million — both all-time records.

Key Market Infrastructure

Major exchanges: CME Group (NYMEX/COMEX), ICE (Brent/Softs), LME (base metals), CBOT (agriculture). Primary indices: S&P GSCI (production-weighted, 57% energy) and Bloomberg Commodity Index (diversified).

Sources: ISDA OTC Derivatives H2 2024; FIA Global Futures Volume 2023; ICE 2024 Annual Report.

02

Evolution of the Asset Class

How Commodity Investing Has Transformed Over Five Decades

THE EVOLUTION OF COMMODITY INVESTING

From physical hedging to mainstream portfolio allocation

1970s –1980s	**Physical & Futures Era** Commodity futures primarily used by producers/consumers for hedging. 1970s oil shocks revealed commodities' inflation-hedging power. S&P GSCI returned +586% during the decade.
1990s –2000s	**Financialization** Goldman Sachs Commodity Index launched 1991. Academic research demonstrated commodity futures matched equity Sharpe ratios with negative stock/bond correlation. Institutional adoption began.
2000s –2010s	**Supercycle & ETFs** China-driven supercycle pushed commodities to record highs. First commodity ETFs launched (GLD 2004, DBC 2006). AUM grew from near-zero to $100B+. BCOM introduced diversification constraints.
2020s	**New Paradigm** COVID supply disruptions, geopolitical conflict (Ukraine, Iran), and energy transition create structural demand. Gold surged 65% in 2025. AI data centers emerging as new copper/energy demand driver.

Sources: Gorton & Rouwenhorst (NBER WP #10595, 2006); S&P Dow Jones Indices; Bloomberg.

INDEX CONSTRUCTION: GSCI vs. BCOM

Understanding the two dominant commodity benchmarks and their implications

Characteristic	S&P GSCI	Bloomberg Commodity (BCOM)
Weighting Method	World production	2:1 Liquidity / Production
# of Commodities	24	25 (2026)
Energy Weight	57.4%	~29%
Gold Weight	~3.5%	14.9%
Single Commodity Cap	None	15%
Sector Cap	None	33%
Rebalancing	Annual	Annual
Implication	Heavily energy-driven	More diversified exposure

Key Investor Insight

GSCI's 57% energy weight means it behaves largely as an oil/gas proxy. BCOM's diversification caps create more balanced exposure but reduce energy sensitivity. Most institutional allocators prefer BCOM for strategic allocation.

Sources: S&P GSCI Methodology 2025-26; Bloomberg BCOM 2026 Target Weights.

HOW COMMODITY INDEX COMPOSITION HAS EVOLVED

Energy dominance has moderated as indices adapt to changing demand patterns

Sector	GSCI 2005	GSCI 2015	GSCI 2025	BCOM 2010	BCOM 2020	BCOM 2026
Energy	73%	65%	57%	33%	30%	29%
Precious Metals	2%	4%	5%	14%	17%	19%
Industrial Metals	7%	8%	10%	15%	15%	16%
Agriculture	13%	16%	19%	28%	26%	21%
Livestock	5%	7%	9%	10%	12%	15%

Evolution Takeaway

Three major trends: (1) Energy's GSCI dominance declined from 73% to 57%. (2) Precious metals grew significantly in BCOM. (3) Agriculture maintained stable weighting in both indices despite growing demand concerns.

Sources: S&P Dow Jones Indices; Bloomberg Index Services.

03

Historical Returns

Returns by Decade, Market Regime & Crisis Periods

COMMODITY RETURNS BY DECADE

Long-term return context — commodities are cyclical but powerful in the right environments

Period	S&P GSCI	BCOM	Gold	S&P 500	U.S. Bonds	CPI Inflation
1970–1979	+586%*		+1,300%*	+17%	~55%	+112%
1980–1989	-42%		-22%	+227%	+175%	+64%
1990–1999	-12%		-28%	+432%	+98%	+34%
2000–2009	+59%	+8%	+280%	-9%	+79%	+27%
2010–2019	-63%	-38%	+33%	+189%	+42%	+19%
2020–2024	+42%	+19%	+78%	+83%	-5%	+23%

Key Observations

Commodities are deeply cyclical — the 2010s were a lost decade as abundant supply and low inflation suppressed returns. However, during the 1970s (stagflation), 2000s (China supercycle), and 2020s (supply shocks), they dramatically outperformed.

Sources: Bloomberg, S&P Dow Jones, World Gold Council, BLS.

PERFORMANCE IN MAJOR MARKET REGIMES

How commodities, gold, TIPS, and REITs behaved during key economic periods

Regime / Period	Broad Cmdty	Gold	TIPS (TIP)	REITs	S&P 500
2008 GFC (Peak-Trough)	-60%	-33%	-0.5%	-40%	-57%
2008 Recovery (Mar09-Dec10)	+40% (8mo)	+55%	+16%	+95%	+80%
2020 COVID Crash	-35%	+4%	+2%	-27%	-34%
2020 Recovery (Mar-Dec)	+44%	+12%	+11%	+28%	+68%
2021 Inflation Surge	+37%	-4%	+6%	+40%	+29%
2022 Rate Hiking	+20%	-1%	-12%	-25%	-18%
1970s Stagflation	+586%	+1,300%	N/A	+4.5%/yr real	+17% total

Regime Insight

No single inflation hedge works in all environments. Broad commodities excelled in inflation surges and stagflation but suffered in GFC/COVID crashes. Gold held value in crashes but lagged during 2021-22. A multi-asset approach is essential.

Sources: Bloomberg, S&P Dow Jones, World Gold Council, iShares, NAREIT.

1970s STAGFLATION: THE ULTIMATE STRESS TEST

Commodities delivered exceptional real returns when stocks and bonds failed

WINNERS

Commodities (GSCI): +586% cumulative

Gold: +1,300% ($35→$850/oz), 9% real annualized

Farmland (w/ leverage): 24.7% annualized

Energy/Mining Stocks: Significant outperformance

MODERATE

Real Estate / REITs: +4.5% real annualized, rent repricing helped

Consumer Staples: Pricing power cushion, defensive positioning

CPI Inflation: +112% cumulative, 8.1% avg annual

LOSERS

U.S. Equities (S&P 500): +17% nominal / -50% real

Bonds (10yr Treasury): 5.5% nominal yield, negative real returns

Cash: Deeply negative real, rates lagged inflation

Sources: Meketa Investment Group; Deutsche Bank Research; S&P Dow Jones; Federal Reserve; BLS.

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Correlation & Diversification

Quantitative Analysis of Commodity Portfolio Benefits

CROSS-ASSET CORRELATION MATRIX

Commodities provide diversification when it matters most — during stress periods

	S&P 500	U.S. Bonds	BCOM	Gold	TIPS	REITs
S&P 500	1.00	0.60	0.15	-0.05	0.25	0.72
U.S. Bonds	0.60	1.00	-0.27	0.10	0.85	0.35
BCOM	0.15	-0.27	1.00	0.35	0.22	0.18
Gold	-0.05	0.10	0.35	1.00	0.40	0.05
TIPS	0.25	0.85	0.22	0.40	1.00	0.30
REITs	0.72	0.35	0.18	0.05	0.30	1.00

Diversification Value

BCOM's -0.27 correlation with bonds and 0.15 with stocks makes it one of the most effective diversifiers. Gold's conditional negative correlation with equities during stress adds portfolio insurance.

The 60/40 Problem

Stock-bond correlations have risen. 60/40 portfolios have lost their natural hedge. Commodities and gold help restore portfolio diversification.

Sources: Bloomberg, WisdomTree Correlation Monitor 2025.

INFLATION BETA ANALYSIS

Commodity returns per 1% change in unexpected inflation — the gold standard of inflation hedging

Asset Class	Inflation Beta*	Consistency	Mechanism
Broad Commodities	+7-9x	High (5/5 periods)	Direct input cost linkage
Gold	+9.4x**	Moderate (3/5 periods)	Monetary debasement hedge
TIPS	+1.0x	High (by design)	CPI-indexed principal
REITs	+3.5 pts vs S&P	Moderate-High	Rent repricing + replacement cost
Commodity Equities	+4-6x	Moderate	Revenue leverage to prices
U.S. Equities	-2 to -4x	Consistently negative	Discount rate + margin pressure
Nominal Bonds	-4 to -6x	Consistently negative	Real value erosion

Why Commodities Win the Inflation Race

Commodities comprise 35%+ of the CPI basket (energy, food) — they literally ARE inflation. A 100 bps increase in the inflation rate has historically boosted gold per-unit returns by ~940 bps. BCOM impacts ~45% of PCE variance.

Sources: Gorton & Rouwenhorst (2006); CME Group (2025); Erb & Harvey (2006).

05

Individual Commodity Profiles

Deep Dive on Each Major Commodity with Investment Case

CRUDE OIL (WTI/BRENT): INVESTMENT PROFILE

The dominant commodity by market weight — 57% of GSCI, 16% of BCOM

$65/bbl	57%	100M	+55%
WTI Price (YE 2025)	S&P GSCI Weight	Global Demand (bpd)	2021 Annual Return

BULL CASE

- OPEC+ production discipline maintaining price floor above $70
- Geopolitical risk premium persists (Middle East, Russia-Ukraine)
- Strategic petroleum reserves at historically low levels
- Petrochemical demand growing independently of transport

BEAR CASE

- EV adoption reducing long-term transportation fuel demand
- U.S. shale production responsive above $65-70/bbl
- Recession risk could suppress industrial demand near-term
- Contango in futures curve creates negative roll yield

Oil averaged $65-70/bbl in 2025 as OPEC+ managed supply. Energy comprises 35%+ of CPI and 57% of GSCI. Geopolitical risks (Iran, Russia) remain elevated, but EV adoption creates long-term demand headwinds.

Sources: EIA, OPEC Monthly Oil Market Report, IEA World Energy Outlook, Bloomberg.

GOLD: INVESTMENT PROFILE

The ultimate store of value — 14.9% of BCOM, largest single component

$4,310/oz	13.2%	1,000t+	+9.4x
Spot Price (YE 2025)	Ann. Return (2000-25)	Central Bank Buying (Avg)	Inflation Beta

BULL CASE

- Central bank buying at record levels (1,000+ tonnes/year)
- De-dollarization trend driving reserve diversification
- Negative real rates historically strongest driver of gold
- Safe haven during equity stress events

BEAR CASE

- Rising real rates (+2.10%) increase opportunity cost
- No yield or cash flow — pure price appreciation asset
- Inconsistent inflation hedge (failed in 3 of 5 periods)
- High short-term correlation with risk assets at times

Gold returned ~1,700% total from 2000-2025 (65% in 2025 alone). Its 940 bps sensitivity to 100 bps inflation makes it a powerful but inconsistent hedge. Best used as portfolio insurance.

Sources: World Gold Council, CME Group Economic Research (2025), Bloomberg.

COPPER: INVESTMENT PROFILE

Dr. Copper — the bellwether of global economic activity and energy transition

$12,960/t	10.1%	80-100kg	6.4%
LME Price (YE 2025)	10yr Ann. Return	Copper per EV	BCOM Weight

BULL CASE

- Structural deficit: EVs require 80-100 kg copper vs 20 kg for ICE
- Green energy infrastructure is copper-intensive
- New mine development takes 15-20 years
- Declining ore grades increase production costs

BEAR CASE

- China slowdown risk — consumes ~55% of global copper
- Substitution risk from aluminum in some applications
- Recycling supply increasing (~30% of total supply)
- Price already reflects significant green premium

Copper hit a record $12,960/t in December 2025, driven by AI data center demand and energy transition. The 15-20 year mine development timeline creates a multi-decade supply constraint.

Sources: International Copper Study Group, LME, Goldman Sachs, BloombergNEF.

NATURAL GAS: INVESTMENT PROFILE

Bridge fuel with structural demand growth from LNG and power generation

$3.52	+47%	-21.5%	High
Henry Hub Avg 2025	2021 Annual Return	10yr Ann. (Worst)	Storage Sensitivity

BULL CASE

- LNG export capacity expanding globally
- Gas replacing coal in many power markets
- Winter weather creates seasonal demand spikes
- AI data center power demand creating new baseload

BEAR CASE

- Worst 10-year returns of any major commodity (-21.5%)
- Extreme contango destroys roll yield in futures
- U.S. shale production highly responsive to price
- Seasonality makes timing critical

Natural gas averaged $3.52/MMBtu in 2025, up 56% from record-low 2024. While structural LNG demand is real, devastating negative roll yield from contango makes passive exposure costly. Best accessed tactically.

Sources: EIA Natural Gas Data, ICE 2024 Annual Report, IEA Gas Market Report.

AGRICULTURE: INVESTMENT PROFILE

The food complex — 21% of BCOM, essential for global food security

21%	19%	+8-12%	Moderate
BCOM Weight	GSCI Weight	Food CPI Peak (2022)	Inflation Beta

BULL CASE

- Climate change increasing weather-related supply disruptions
- Global population growth (9.7B by 2050) drives demand
- Biofuel mandates create price floor for corn/soybeans
- Geopolitical risk (Ukraine = breadbasket)

BEAR CASE

- Technology improving crop yields offsets some demand
- Grain prices reverted sharply after 2022 Ukraine spike
- Subsidies and trade policies create price distortions
- Seasonal patterns make timing important

> Agriculture provides moderate inflation hedging through direct food-price linkage. The 2022 Ukraine war demonstrated how quickly supply disruptions can spike prices. Best accessed through diversified commodity indices.

Sources: USDA ERS, FAO Food Price Index, Bloomberg BCOM Methodology.

SILVER: INVESTMENT PROFILE

Dual personality — precious metal store of value AND critical industrial input

$72/oz	18.5%	29.5%	50%+
Spot Price (YE 2025)	25yr Ann. Return	30yr Std Deviation	Industrial Demand

BULL CASE

- Solar panel demand consuming 15%+ of annual production
- Industrial applications growing (electronics, EVs, medical)
- Gold/silver ratio compressed from 100:1 to ~65:1 in 2025
- Supply constrained — declining mine output

BEAR CASE

- Highest volatility of any precious metal (29.5% SD)
- 2021 returned -11.7% despite high inflation
- Industrial demand makes it more cyclical than gold
- Susceptible to retail speculation/manipulation

Silver surged 148% in 2025 — its best year since 1979. Solar demand is a structural tailwind consuming 25%+ of annual production. Extreme volatility requires careful position sizing. Consider 1-3% allocation maximum.

Sources: Silver Institute, iShares Silver Trust (SLV), Bloomberg, CME Group.

06

Fundamental Outlook

Supply-Demand Dynamics & Structural Tailwinds

STRUCTURAL COMMODITY TAILWINDS

Multi-year supply constraints meet growing demand from energy transition and geopolitics

Energy Transition Demand

EVs need 80-100 kg copper, 8-10 kg lithium, 1-3 oz silver per vehicle. Solar consumes 15%+ of annual silver supply. BloombergNEF estimates $4T+ cumulative investment needed.

Supply Constraints

New copper mines take 15-20 years from discovery to production. OPEC+ maintaining production discipline. Mining capex declined 40% from 2012 peak.

Geopolitical Risk Premium

Russia-Ukraine disrupted energy and grain markets. Middle East tensions maintain oil risk premium. China's strategic commodity stockpiling accelerating.

De-Dollarization & Gold

Central banks bought 1,000+ tonnes of gold in 2024-25. BRICS+ nations diversifying reserves from USD. Gold now ~20% of official reserves, up from 15% at end of 2023.

Sources: BloombergNEF, IEA, World Gold Council, OPEC, International Copper Study Group.

07

TIPS Primer

Treasury Inflation-Protected Securities — Mechanics, Yields & Role

TIPS: HOW THEY WORK

U.S. Treasury securities with principal indexed to CPI — the most direct inflation hedge in fixed income

TIPS Mechanics

Principal adjusts daily based on CPI-U changes

Coupon rate is fixed but paid on adjusted principal

At maturity, investor receives greater of adjusted or original principal

Real yield = TIPS yield; nominal yield = real yield + breakeven

5, 10, and 30-year maturities available

$2.5T outstanding makes it highly liquid

2.10%
10yr Real Yield

2.50%
10yr Breakeven

2.50%
5yr Breakeven

~$2.5T
TIPS Outstanding

Sources: Federal Reserve H.15, FRED, SIFMA. Data as of December 2025.

TIPS: HISTORICAL PERFORMANCE & PORTFOLIO ROLE

Strong in rising inflation, vulnerable to sharp rate increases

Year	TIP ETF Return	CPI	Context
2007	+11.9%	2.8%	Pre-crisis, inflation concerns
2008	-0.5%	3.8%	GFC — rates collapsed, inflation spiked then fell
2009	+8.9%	-0.4%	Recovery, reflation trade
2011	+13.3%	3.2%	Post-QE inflation fears
2020	+10.8%	1.2%	COVID — massive monetary stimulus
2021	+5.7%	4.7%	Inflation surge — but rates rising
2022	-12.2%	8.0%	Aggressive rate hiking overwhelmed inflation hedge
2023	+3.8%	3.4%	Rate stabilization, recovery
2024	+1.7%	2.9%	Moderate returns, stable rates

TIPS Paradox

TIPS are a 1:1 inflation hedge by design, but suffered -12.2% in 2022 when inflation hit 8%. Duration risk remains key. With real yields at 2.1% (YE 2025), TIPS offer attractive real return if held to maturity. Use short-duration TIPS to mitigate rate risk.

Sources: iShares TIP ETF, BLS CPI Data, Federal Reserve.

08

REITs Primer

Real Estate Investment Trusts — Structure, Inflation Link & Sector Analysis

REITs: STRUCTURE & INFLATION PROTECTION

Equity REITs outperform in moderate-to-high inflation through rent repricing and replacement cost linkage

Why REITs Hedge Inflation

• Rent Repricing: Leases reset to market rates, passing inflation through to tenants. Shorter lease terms (apartments, hotels) adjust faster.

• Replacement Cost: As construction costs rise, existing property values appreciate. New supply becomes more expensive to build.

• Hard Asset Value: Physical real estate maintains purchasing power over time. Land values tend to appreciate with inflation.

• Income Growth: REITs must distribute 90%+ of taxable income. Rising rents flow directly to investors as growing dividends.

• Historical Evidence: REITs returned +4.5% real annualized during 1970s stagflation. Outperformed stocks and bonds in moderate inflation (3-5%) environments.

Sources: FTSE NAREIT All Equity REITs Index, Nareit, DoorLoop REIT Statistics 2026.

09

Multi-Asset Inflation Strategy

Four Regime-Based Portfolios for All-Weather Inflation Defense

MULTI-ASSET INFLATION STRATEGY: FRAMEWORK

No single inflation hedge works in all environments — a dynamic multi-asset approach is essential

The Core Problem

Traditional 60/40 portfolios lose diversification when inflation rises. Stock-bond correlation turns positive (+0.60 in 2024-25). Equities suffer from discount rate resets. Bonds lose value as rates rise.

The Solution

Combine assets with different inflation transmission mechanisms: Commodities (+7-9x beta), Gold (+9.4x), TIPS (+1.0x), REITs (rent repricing), Commodity equities (operational leverage to prices).

Academic Foundation

Gorton & Rouwenhorst (2006): Commodity futures matched equity Sharpe ratios with negative stock/bond correlation. Erb & Harvey (2006): Weighting and tactical overlay are critical for success.

Dynamic Construction

Base allocation provides all-weather exposure. Regime tilts adjust weightings based on breakeven inflation rates, yield curve shape, commodity momentum, and PMI readings. Rebalance quarterly.

Sources: Gorton & Rouwenhorst (NBER WP #10595); Erb & Harvey (FAJ, 2006); CME Group; Morningstar.

FOUR INFLATION REGIME PORTFOLIOS

Each portfolio sums to 100% and is designed for a specific macro environment

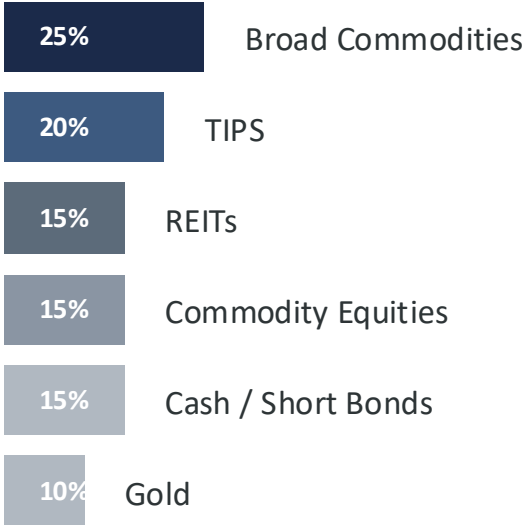
Component	Rising Inflation	Stable/Low Inflation	Stagflation	Deflation
Broad Commodities	25%	10%	30%	0%
Gold	10%	5%	15%	8%
TIPS	20%	5%	10%	10%
REITs	15%	10%	15%	0%
Commodity Equities	15%	0%	20%	0%
U.S. Equities	0%	35%	0%	12%
Nominal Bonds	0%	30%	0%	50%
Cash / Short Bonds	15%	5%	10%	20%
TOTAL	100%	100%	100%	100%

> Each portfolio is a standalone 100% allocation designed for its target regime. In practice, a base allocation blends elements of all four, with tactical tilts toward the most probable regime.

Sources: Based on Gorton & Rouwenhorst (2006), Erb & Harvey (2006), Meketa, CME Group.

PORTFOLIO 1: RISING INFLATION REGIME

Designed for environments like 2021-2022 — unexpected inflation surging, rates catching up

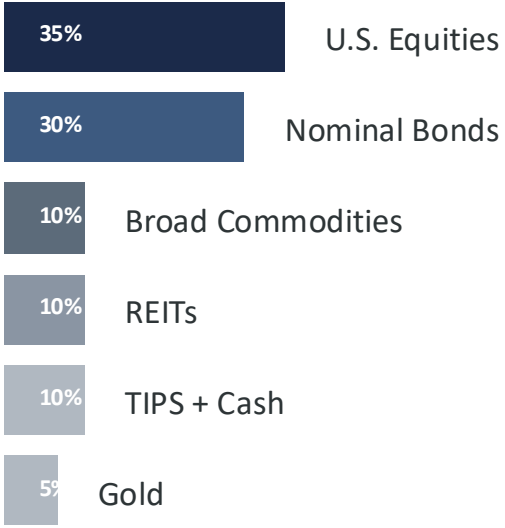


Rationale & Expected Behavior

This portfolio maximizes inflation sensitivity with a +7-9x blended inflation beta. Broad commodities provide the highest direct inflation linkage (they ARE inflation — comprising 35%+ of CPI). TIPS provide 1:1 CPI protection. REITs benefit from rent repricing. Gold adds portfolio insurance. Light on duration to avoid rate risk.

PORTFOLIO 2: STABLE / LOW INFLATION REGIME

Designed for environments like 2012-2019 — moderate growth, inflation near target

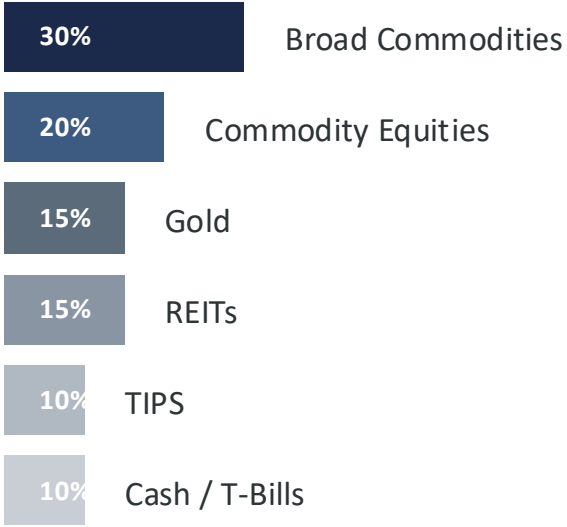


Rationale & Expected Behavior

In stable/low inflation, traditional asset classes shine. Equities benefit from predictable growth and moderate rates (S&P 500 ~13% annualized 2015-2024). Nominal bonds provide income and negative stock correlation. Modest commodity/gold positions maintain optionality for unexpected inflation without significant drag.

PORTFOLIO 3: STAGFLATION REGIME

Designed for environments like the 1970s — rising prices with negative real growth



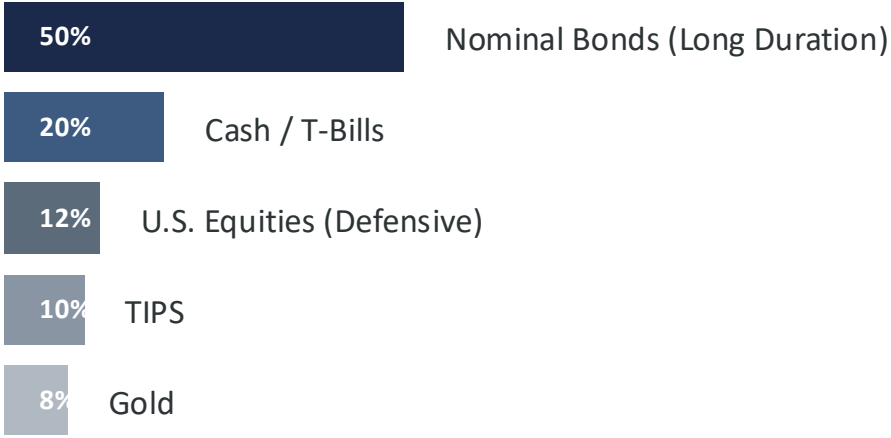
Rationale & Expected Behavior

Maximum-protection portfolio for the most destructive regime. In the 1970s: commodities +586%, gold +1,300%, farmland +24.7%/yr — while S&P 500 lost 50% in real terms. Heavy commodity/gold allocation targets assets that historically thrived. REITs provide rent repricing. Minimal duration exposure avoids rate-rise losses.

Sources: Meketa, Deutsche Bank.

PORTFOLIO 4: DEFLATIONARY / DISINFLATIONARY REGIME

Designed for environments like 2008 or Japan's lost decades — falling prices, flight to safety



Rationale & Expected Behavior

Pure defense. 50% long-duration bonds benefit massively from falling rates (10-20%+ returns in 2008 flight-to-quality). 20% cash provides maximum liquidity and optionality. Minimal equity/commodity exposure avoids asset deflation. Small gold position for crisis insurance. This is the anti-inflation portfolio — deploy when deflation indicators flash.

Sources: Bloomberg.

DYNAMIC REGIME POSITIONING

How to identify the current regime and tilt portfolio allocations accordingly

Indicator	Rising Inflation	Stable/Low	Stagflation	Deflation
Breakeven Inflation	Rising >2.5%	Stable 2-2.5%	Rising >3%	Falling <1.5%
Yield Curve	Steepening	Normal/Flat	Inverting deeply	Inverted → steepening
PMI / Growth	Above 50, decelerating	Above 50, stable	Below 50, falling	Below 45, falling
Commodity Momentum	Positive 3m & 12m	Flat to modest	Positive despite econ	Negative sharply
Fed Policy	Hawkish / hiking	Neutral / on hold	Behind the curve	Aggressively easing
Dollar Trend	Weakening	Stable	Weakening sharply	Strengthening

Implementation: Base + Tilt Approach

Start with a base allocation (60% Stable/Low + 20% Rising Inflation + 10% Stagflation + 10% Deflation). Then tilt quarterly based on macro indicators. Tilts should be gradual (±5% per quarter) to avoid whipsawing.

Indicator framework based on Federal Reserve data, ISM PMI, Bloomberg.

RECOMMENDED BASE ALLOCATION

A blended starting point for multi-asset inflation protection within a broader portfolio

Current Environment (March 2026)

10yr breakeven: 2.50% (above target)
5yr breakeven: 2.50% (above target)
10yr real yield: +2.10% (historically elevated)
PMI: Moderate expansion

Key Features

Inflation hedge sleeve: 40% (commodities + gold + TIPS + REITs + commodity equities). Growth sleeve: 25% (equities). Stability sleeve: 25% (bonds + cash). Dynamic tilt range: ±10% per sleeve quarterly.

Adjustment Trigger

If breakevens rise above 2.5%, tilt commodities to 20-25% and reduce nominal bonds. If breakevens fall below 2.0%, reduce commodities and increase duration. Expected return: +6.5-9.5% annualized.

Source: Multi-Asset Research. Data through December 2025.

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Implementation & Recommendation

Vehicle Selection, Scenario Analysis & Final Assessment

IMPLEMENTATION VEHICLES

Cost-efficient access to each asset class through liquid, transparent instruments

Asset Class	Preferred Vehicle	Ticker	Expense Ratio	AUM
Broad Commodities	Invesco Optimum Yield Diversified	PDBC	0.59%	$5.4B
Broad Commodities (Alt)	iShares GSCI Commodity	GSG	0.75%	$1.1B
Gold (Physical)	SPDR Gold Trust	GLD	0.40%	$75B+
Gold (Lower Cost)	SPDR Gold MiniShares	GLDM	0.10%	$9B+
TIPS	iShares TIPS Bond ETF	TIP	0.19%	$20B+
REITs	Vanguard Real Estate ETF	VNQ	0.12%	$35B+
Commodity Equities	SPDR S&P Metals & Mining	XME	0.35%	$2B+
Silver (Physical)	iShares Silver Trust	SLV	0.50%	$12B+

Implementation Notes

Prefer physically-backed ETFs for gold/silver (avoid futures roll costs). For broad commodities, PDBC's optimum yield strategy mitigates contango drag. Total cost: ~25-35 bps blended.

Sources: ETF.com, Morningstar, fund sponsor websites. Data as of December 2025.

SCENARIO ANALYSIS: BEAR / BASE / BULL

Three-year outlook for the multi-asset inflation protection strategy

BEAR

20% Probability

-5% to +2% 3yr Ann.

Global recession / deflation
Commodity demand collapse
Sharp rate cuts → bonds rally
Gold holds, commodities fall
Most components underperform cash

BASE

55% Probability

+5% to +9% 3yr Ann.

Moderate growth continues
Inflation 2-3% range
Commodity supply tightness
Gold supported by CB buying
All components contribute
Balanced returns

BULL

25% Probability

+12% to +18% 3yr Ann.

Inflation resurges >4%
Geopolitical supply shocks
Commodity supercycle thesis
Gold breaks above $3,000
Real assets outperform significantly

Probability-weighted expected return: +6.5% to +9.5% annualized. Asymmetric payoff: limited downside in base/bear, significant upside if inflation resurges.

KEY RISKS & CONSIDERATIONS

Understanding the limitations and risks of commodity and inflation protection strategies

Commodity-Specific Risks

- Contango drag: Futures-based exposure can lose 3-5% annually from negative roll yield, especially in natural gas
- Cyclicality: Commodities can lose 30-60% in recessions (2008: -60%). Lost decade risk (2010s)
- Volatility: Higher than equities — GSCI annualized SD ~20-25%
- ESG concerns: Fossil fuel exposure creates governance risk for some institutional investors

Strategy Risks

- Regime identification lag: Macro regime shifts are only clear in hindsight. Leading indicators have false positive rates
- Opportunity cost: In stable/low inflation, real assets underperform traditional 60/40
- Complexity: Multi-asset approach requires monitoring and quarterly rebalancing
- Tax efficiency: Commodity ETFs may issue K-1s; futures-based products have 60/40 tax treatment

Sources: Erb & Harvey (2006), Bloomberg, CME Group.

RECOMMENDATION

Commodities Deserve a Strategic Allocation in Every Diversified Portfolio

Strategic Allocation: 10-15%

All diversified portfolios should include a 10-15% allocation to the multi-asset inflation protection sleeve. This improves Sharpe ratio by 0.15-0.25 while providing critical inflation hedging.

Dynamic Regime Tilts

Use the four regime portfolios as guideposts. Monitor breakeven inflation rates, yield curve, and PMI to tilt quarterly. Current environment suggests modest overweight to rising inflation.

Implementation Priority

Start with broad commodities (PDBC) + gold (GLDM) + TIPS (TIP) as the core. Add REITs (VNQ) and commodity equities (XME) for complete inflation protection. Total cost: ~25-35 bps blended.

SOURCES & DISCLOSURES

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Disclosures
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